

Decision Problems, Recurrence Sequences, and Continued Fractions

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Abstract

These notes contain material from the ‘chalkboard’ segments of a mini-course I gave at the Institute of Mathematics of the Polish Academy of Sciences as part of the Simons Semester: “Continued fractions, fractals, ergodic theory and dynamics.”

1 Lecture 1: Positivity for C-finite Sequences

The material in this lecture follows closely (arguably *identically*) the presentation in [OW15].

1.1 Preliminaries for C-finite sequences

A real-valued sequence $\langle u_n \rangle_{n=0}^\infty$ is said *C-finite* if it satisfies a recurrence relation of the form

$$u_n = a_1 u_{n-1} + \cdots + a_k u_{n-k}, \quad (a_k \neq 0)$$

If the preceding relation is the shortest such relation satisfied by $\langle u_n \rangle_n$, then we say that $\langle u_n \rangle_n$ has *order* k . The associated *characteristic polynomial* is given by $p(x) = x^k - a_1 x^{k-1} - \cdots - a_{k-1} x - a_k$. The roots of p are called the *characteristic roots* of $\langle u_n \rangle_n$ and its dominant roots are those of maximal modulus.

We consider real-valued C-finite sequences and so the associated characteristic polynomials have real coefficients and non-real roots always appear in complex conjugate pairs. Thus the set of characteristic roots is of the form $\{\rho_1, \dots, \rho_\ell, \lambda_1, \overline{\lambda_1}, \dots, \lambda_m, \overline{\lambda_m}\}$ with each $\rho_i \in \mathbb{R}$ and each $\lambda_i \in \mathbb{C} \setminus \mathbb{R}$.

Each C-finite sequence $\langle u_n \rangle_n$ admits an exponential-polynomial form. That is to say, there exist univariate polynomials $A_1, \dots, A_\ell$ and $C_1, \dots, C_m$ such that for all $n \in \mathbb{N}_0$,

$$u_n = \sum_{i=1}^{\ell} A_i(n) \rho_i^n + \sum_{j=1}^m (C_j(n) \lambda_j + \overline{C_j(n)} \overline{\lambda_j}).$$

For integer-valued C-finite sequences, the polynomial coefficients A_i have real algebraic coefficients, whilst the polynomials C_j have complex algebraic coefficients. Moreover, the degrees of each of the polynomials is at most $d - 1$, where d is the multiplicity of the corresponding characteristic root.

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1.2 The Positivity Problem

POSITIVITY asks, given a recursively defined real-valued sequence $\langle u_n \rangle_n$, is $u_n \geq 0$ for all n ?

In what is quite clearly a crime against terminology, a sequence is *positive* if its terms are all non-negative.

The structure of the roots of the characteristic polynomial is of paramount importance. The following classical result implies that one need only consider C-finite sequences with real positive characteristic roots in order to decide POSITIVITY in this setting.

Theorem 1 (Pringsheim). *Let $\langle u_n \rangle_{n=0}^\infty$ be a C-finite sequence with no real positive characteristic roots. Then there are infinitely many $n \in \mathbb{N}_0$ for which $u_n < 0$ and, similarly, there are infinitely many $n \in \mathbb{N}_0$ for which $u_n > 0$.*

1.3 A worked example

This subsection focuses on exposition and the material herein, specifically the worked example, is taken from [OW15].

Example 2. This example is taken wholesale from [OW15].

Let $\langle u_n \rangle_n$ be the C-finite sequence defined as follows. The n th term is given by

$$u_n = \frac{33}{8} + \lambda_1^n + \overline{\lambda_1^n} + 2\lambda_2^n + 2\overline{\lambda_2^n} \quad \text{where } \lambda_1 := \frac{-3 + 4i}{5} \text{ and } \lambda_2 := \frac{-7 + 24i}{25}.$$

Our goal in this worked example is to determine whether $\langle u_n \rangle_n$ is positive. Observe that $\langle u_n \rangle_n$ is a sequence of order 5, all of its roots lie on the unit circle $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$, and neither λ_1 nor λ_2 is a root of unity. It follows that $\{\lambda_1^n : n \in \mathbb{N}\}$ and $\{\lambda_2^n : n \in \mathbb{N}\}$ are both dense on $\mathbb{T}$.

A sufficient condition for $\langle u_n \rangle_n$ to fail the positivity test is as follows, is it possible that there is an $n \in \mathbb{N}$ such that both $\operatorname{Re}(\lambda_1^n)$ and $\operatorname{Re}(\lambda_2^n)$ are at most $-3/4$?¹ (Such a condition certainly holds if the orbit $\{(\lambda_1^n, \lambda_2^n) : n \in \mathbb{N}\}$ is a dense subset of $\mathbb{T}^2$).

However, such a condition cannot hold. We reason as follows. First, characteristic roots obey the multiplicative relation $\lambda_1^n \lambda_2^n = 1$. Second, $\{(\lambda_1^n, \lambda_2^n) : n \in \mathbb{N}\} \subset \{(z_1, z_2) \in \mathbb{T}^2 : z_1^2 z_2 = 1\} =: \mathbf{T}$ and further, $\{(n_1, n_2) \in \mathbb{Z}^2 : \lambda_1^{n_1} \lambda_2^{n_2} = 1\}$ is a rank-1 subgroup of $\mathbb{Z}^2$ with generator $(2, 1)$. By Kronecker's Theorem [Cas57, pg.53], $\{(\lambda_1^n, \lambda_2^n) : n \in \mathbb{N}\}$ is a dense subset of $\mathbf{T}$.

We now associate to the discrete sequence $\langle u_n \rangle_n$ a continuous function $f : \mathbb{T}^2 \rightarrow \mathbb{R}$ given by $f(z_1, z_2) = z_1 + \overline{z_1} + 2z_2 + 2\overline{z_2}$. Then $u_n = f(\lambda_1^n, \lambda_2^n)$ and, by calculus, f is non-negative on $\mathbf{T}$. Thus $u_n \geq 0$ for all $n \in \mathbb{N}_0$, and so we determine that $\langle u_n \rangle_n$ is positive, as desired.

Let us extend the current example to consider an order-6 C-finite sequence $\langle v_n \rangle_n$ with terms given by $v_n := u_n - 1/2^n$. We previously established the lower bound $u_n \geq 0$. To determine whether $\langle v_n \rangle_n$ is positive, it suffices to establish the tighter lower bound $u_n \geq 1/2^n$ for all $n \in \mathbb{N}$. This will require more machinery than before since $\liminf_{n \rightarrow \infty} u_n = 0$. This claim is easily seen to hold by analysing the zeroes of the function f (as above). There are two points (z_1^*, z_2^*) and $(\overline{z_1^*}, \overline{z_2^*})$ in $\mathbf{T}$ where f vanishes. Since $\{(\lambda_1^n, \lambda_2^n) : n \in \mathbb{N}\}$ is a dense subset of $\mathbf{T}$, we have that

$$\liminf_{n \rightarrow \infty} u_n = \liminf_{n \rightarrow \infty} f(\lambda_1^n, \lambda_2^n) = 0,$$

as desired.

We shall make use of the following simplified version of Baker's theorem on linear forms in logarithms of algebraic numbers. Our application is to bound from below $f(\lambda_1^n, \lambda_2^n)$ as a function of n via lower bounds on the distance between $(\lambda_1^n, \lambda_2^n)$ and the two zeroes of f in $\mathbf{T}$.

¹If yes, then $u_n < 33/8 - 3/4 - 3/4 - 6/4 - 6/4 = -3/8 < 0$ and so $\langle u_n \rangle_n$ is not positive.

Theorem 3 ((Simple) Baker for linear forms in 3 logarithms). *Let $\alpha_1, \alpha_2, \alpha_3$ be algebraic numbers and b_1, b_2, b_3 integers of absolute value at most H . Then for the principal branch of the complex logarithm,*

$$\Lambda := b_1 \log \alpha_1 + b_2 \log \alpha_2 + b_3 \log \alpha_3$$

is either 0 or satisfies $|\Lambda| > H^{-C}$ where C is effectively computable and depends only on $\alpha_1, \alpha_2, \alpha_3$.

We now turn our attention to bounding the distance between λ_1^n and z_1^* as follows. Recall that $\log \alpha = i \arg \alpha$ for any $\alpha \in \mathbb{T}$. Let n be a natural number such that $\lambda_1^n \neq z_1^*$ and choose $m \in \mathbb{Z}$ so that $-\pi < n \arg \lambda_1 - 2\pi m \leq \pi$. Then

$$|\lambda_1^n - z_1^*| \geq \frac{1}{2} |n \arg \lambda_1 - 2\pi m - \arg z_1^*| \geq \frac{1}{\text{poly}(n)} \quad (1)$$

where $\text{poly}(n)$ is effectively computable and depends on λ_1 and z_1^* . In the above, the first inequality follows from estimating the distance of the shorter circular arc connecting λ_1 and z_1^* , whilst the second inequality is an application of [Theorem 3](#).

We can now compute an $N \in \mathbb{N}$ such that $u_n \geq 1/2^n$ for all $n > N$. First, let $N_1 \in \mathbb{N}$ be such that for all $n > N_1$, we have that $(\lambda_1^n, \lambda_2^n)$ is neither of the points (z_1^*, z_2^*) and $(\overline{z_1^*}, \overline{z_2^*})$ where f vanishes. Second, let $N_2 \in \mathbb{N}$ be such that for all $n > N_2$ we can employ the lower bound $\text{poly}(n)$. Third, let $N_3 \in \mathbb{N}$ be such that for all $n > N_3$ we have that $1/\text{poly}(n) \geq 1/2^n$. Finally, take $N = \max N_1, N_2, N_3 + 1$. Then $u_n \geq 1/2^n$ for all $n > N$, as desired. Thus deciding the positivity of $\langle v_n \rangle_n$ reduces to an exhaustive search where we want to determine whether $v_n \geq 0$ for each $n \in \{0, 1, \dots, N\}$.

We conclude this section by noting that the approach to POSITIVITY in [Example 2](#) does not generalise to higher-order C-finite sequences. Indeed, POSITIVITY is only known to be decidable up to order 5. In fact deciding POSITIVITY at order 6 would allow one to compute the homogeneous Diophantine approximation type of a large class of transcendental numbers (cf. [\[OW15\]](#)). POSITIVITY at higher orders is (obviously) an open decision problem.

In the next lectures we shall abandon the C-finite setting for second-order recursively defined sequences.

2 Lecture 2: Minimality for Second-Order Recurrence Sequences

The material in this lecture follows closely (arguably *identically*) the presentation in [\[LW92\]](#).

2.1 Notation for continued fractions

The continued fraction algorithm is the function $\mathbb{K}$ mapping a pair $(\langle a_n \rangle_n, \langle b_n \rangle_n)$ onto the sequence $\langle f_n \rangle_n$ where

$$f_n := \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \dots + \frac{b_n}{a_n}}}} = \frac{A_n}{B_n}$$

where $\langle A_n \rangle_n$ and $\langle B_n \rangle_n$, the respective *canonical numerators* and *denominators* satisfy the recurrence relations

$$\begin{aligned} A_n &= b_n A_{n-1} + a_n A_{n-2}, \quad \text{and} \\ B_n &= b_n B_{n-1} + a_n B_{n-2} \end{aligned}$$

for each $n \in \mathbb{N}$ with initial conditions $A_{-1} = 1, A_0 = 0$, and $B_{-1} = 0, B_0 = 1$.

2.2 Structure of second-order solution spaces

Consider the general second-order recurrence relation

$$u_n = b_n u_{n-1} + a_n u_{n-2} \quad \text{for } n \in \mathbb{N} \quad (2)$$

where all the a_n and b_n are complex numbers $a_n \neq 0$ for all $n \in \mathbb{N}$.

A sequence $\langle v_n \rangle_{n=-1}^\infty$ is a *solution* of (2) if its elements satisfy the relation for each $n \in \mathbb{N}$. A solution $\langle v_n \rangle_n$ of (2) is uniquely determined by the pair of initial values v_{-1} and v_0 . The set of all solutions of (2) is the relation's *solution space*. The solution space has the following nice properties:

1. the sequence $\langle 0 \rangle_n$ is a solution (henceforth the *trivial solution*,
2. if $\langle v_n \rangle_n$ is a solution, then so is $\alpha \langle v_n \rangle_n = \langle \alpha v_n \rangle_n$ for every $\alpha \in \mathbb{C}$.
3. if $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$ are solutions, then so is $\langle w_n \rangle_n + \langle v_n \rangle_n = \langle v_n + w_n \rangle_n$.

Theorem 4 ([LW92, pg.192]). *The solution space of the second-order recurrence relation (2) is a vector space of dimension 2. The canonical numerators and denominators, $\langle A_n \rangle_n$ and $\langle B_n \rangle_n$, of $\mathbf{K}_{n=1}^\infty(a_n/b_n)$ form a basis for the solution space.*

Example 5. The sequences with terms $\left(\frac{1-\sqrt{5}}{2}\right)^{n+1}$ and $\left(\frac{1+\sqrt{5}}{2}\right)^{n+1}$ are easily checked to be linearly independent solutions of the second-order Fibonacci relation.

Let $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$ be two linearly independent solutions of the same recurrence relation (2). Then there is no $n \in \{-1, 0, 1, \dots\}$ for which w_n and v_n are both 0. This follows from the classical determinant property of such sequences (cf. [LW92, pg.196]). Thus w_n/v_n is well-defined (in $\hat{\mathbb{C}}$) for each n .

2.3 Connecting minimal solutions to the convergence of continued fractions

Definition 6 (Minimal solutions). A solution $\langle u_n \rangle_{n=-1}^\infty$ to (2) is *minimal* if $\langle u_n \rangle_n$ is non-trivial and there exists a second solution $\langle v_n \rangle_n$ to (2) such that $\lim_{n \rightarrow \infty} u_n/v_n = 0$. In such cases, $\langle v_n \rangle_n$ is said to be *dominant*.

Suppose that $\langle u_n \rangle_n$ is a minimal solution of (2) and $\langle v_n \rangle_n$ is a dominant solution of the same recurrence relation (2). We make the following observations:

1. the solution $\alpha \langle u_n \rangle_n$ with $\alpha \neq 0$ is also a minimal,
2. taken together, $\langle u_n \rangle_n$ and $\langle v_n \rangle_n$ form a basis set for the solution space of (2). A non-trivial solution $\langle w_n \rangle_n$ of (2) can be written as $\langle w_n \rangle_n = c_1 \langle u_n \rangle_n + c_2 \langle v_n \rangle_n$ and the condition $c_2 \neq 0$ is equivalent to $\langle w_n \rangle_n$ being a dominant solution.
3. For a non-trivial solution $\langle w_n \rangle_n$ as above, $w_n/v_n \rightarrow c_2 \neq 0$. The converse is also true, if $w_n/v_n \rightarrow c_2 \in \hat{\mathbb{C}} \setminus \{0\}$, then $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$ are both dominant.

We are motivated by the following decision problems:

- Given a recurrence relation of the form (2), does the solution space contain minimal solutions?
- MINIMALITY PROBLEM: Given a second-order sequence $\langle u_n \rangle_n$ satisfying (2), is $\langle u_n \rangle_n$ minimal?

Theorem 7. *Suppose that $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$, are two linearly independent solutions to (2) such that $\lim_{n \rightarrow \infty} w_n/v_n$ exists ($= R \in \hat{\mathbb{C}}$). Then (2) admits minimal solutions.*

Proof. If $R = \infty$, then $\langle v_n \rangle_n$ is minimal. If $R \neq \infty$, then $\langle w_n - Rv_n \rangle_n$ is a non-trivial solution of (2) and

$$\lim_{n \rightarrow \infty} \frac{w_n - Rv_n}{v_n} = 0.$$

Thus $\langle w_n - Rv_n \rangle_n$ is minimal. □

We shall discuss a classical result of Pincherle [Pin94] that connects the existence of minimal solutions of a given recurrence relation to the convergence of the associated continued fraction.

Non-Example 8. Not all second-order recurrence relations admit minimal solutions. Take, for example, the relation $u_n = u_{n-2}$. The associated solution space is spanned by the pair of linearly independent solutions $\langle w_n \rangle_n = \langle 0, 1, 0, 1, \dots \rangle$ and $\langle v_n \rangle_n = \langle 1, 0, 1, 0, \dots \rangle$.

Example 9. Consider $u_n = u_{n-1} + u_{n-2}$, $n = 1, 2, \dots$. All solutions satisfy the Binet formula $u_n = a\phi^n + b(-\phi^{-n})$ where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio and the constants, a and b in the formula, are determined by substituting initial values. We have that

$$\begin{cases} u_{-1} &= a\phi^{-1} + b(-\phi), \\ u_0 &= a + b. \end{cases}$$

In this example (and for all second-order recurrence relations with constant coefficients that admit minimal solutions), it is fairly straightforward to determine if a solution is minimal. Observe that $\lim_{n \rightarrow \infty} u_n/\phi^n = 0$ if and only if $a = 0$. Thus a solution $\langle u_n \rangle_n$ of the Fibonacci recurrence is minimal if and only if $u_0/u_{-1} = b/(-b\phi) = -\phi^{-1}$. In summary, given the initial terms of a solution sequence $\langle u_n \rangle_n$ of the Fibonacci recurrence, to test whether $\langle u_n \rangle_n$ is minimal it is sufficient to test whether we have equality between u_0/u_{-1} and $-\phi^{-1}$.

Example 10. Consider the Apéry recurrence relation

$$u_n = \frac{34n^3 - 51n^2 + 27n - 5}{n^3} u_{n-1} - \frac{(n-1)^3}{n^3} u_{n-2}. \quad (3)$$

Since this recurrence relation has polynomial coefficients, rather than constant coefficients, we can no longer rely of a closed exponential-polynomial form for its solution sequences. However, we do know that the solution space is two-dimensional and, it just so happens, that for this example, we have the following solutions (cf. [Apé79; Poo79]):

$$\begin{aligned} \langle A_n \rangle_{n=0}^\infty &= \langle 1, 5, 73, 1445, \dots \rangle, \\ \langle B_n \rangle_{n=0}^\infty &= \langle 0, 6, 351/4, 62531/36, \dots \rangle. \end{aligned}$$

van der Poorten's exposition [Poo79] demonstrates that $\zeta(3) := \sum_{k=1}^\infty 1/k^3$ can be approximated by the sequence with n th term values B_n/A_n . Omitting many important details, the convergence of $B_n/A_n \rightarrow \zeta(3)$ is so fast as to ensure that $\zeta(3)$ is irrational.

We apply Theorem 7 to the above to find that $\alpha \langle B_n - \zeta(3)A_n \rangle_n$ with $\alpha \neq 0$ is a minimal solution. Thus a solution $\langle u_n \rangle_n$ to Apéry's recurrence relation (3) is minimal if and only if the ratio of its initial values satisfies $u_1/u_0 = (\zeta(3)5 - 6)/\zeta(3)$.

It now apparent that the necessary equality test to determine an instance of MINIMALITY for the Apéry recurrence is harder than the preceding example: we may have a very good grasp on the inputs u_0 and u_1 , but we are at more of a loss when it comes to the constant $(\zeta(3)5 - 6)/\zeta(3)$. Indeed, it is known that $\zeta(3)$ is irrational, but it is open whether $\zeta(3)$ is algebraic or transcendental.

We now turn to the continued fraction connection. We give a brief insight into Pincherle's theorem before giving a formal statement. Recall that the approximants of $\mathbf{K}(a_n/b_n)$ satisfy

$$f_n = \frac{A_n}{B_n} = \frac{\alpha_1 w_n + \alpha_2 v_n}{\beta_1 w_n + \beta_2 v_n} = \frac{\alpha_1 w_n/v_n + \alpha_2}{\beta_1 w_n/v_n + \beta_2}$$

where $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$ are solutions of the recurrence relation (2). Notice that if $w_n/v_n \rightarrow 0$ (of minimality fame), then $\langle f_n \rangle_n$ converges to the limit α_2/β_2 . Pincherle's theorem is discussed in several sources (e.g., [Gau67, pg. 31], [LW92, pg. 202], [MW05, pg. 4]). The textbook [LW92] includes a formal proof of the result.

Theorem 11 (Pincherle [Pin94]). *Let $K(a_n/b_n)$ be a continued fraction with elements $\langle a_n \rangle_n$ and $\langle b_n \rangle_n$ all from $\mathbb{C}$ and $a_n \neq 0$. Then,*

1. $K(a_n/b_n)$ converges in $\hat{\mathbb{C}}$ if and only if the associated recurrence admits minimal solutions.
2. If (2) has a minimal solution then $K(a_n/b_n)$ converges to $-w_0/w_{-1} \in \hat{\mathbb{C}}$.

3 Lecture 3: Positivity and Minimality for Second-Order P-finite Sequences

The exposition and material in this lecture relates to the work in [Ken+21; Ken+].

A rational-valued sequence $\langle u_n \rangle_{n=-1}^\infty$ is a *second-order P-finite sequence* if each of its terms satisfy a recurrence relation of the form

$$p_3(n)u_n = p_2(n)u_{n-1} + p_1(n)u_{n-2} \quad n = 1, 2, 3, \dots \quad (4)$$

where each $p_i \in \mathbb{Q}[x]$, $p_1(n), p_3(n) \neq 0$ for all $n \in \mathbb{N}$. An excellent textbook for relevant background material on P-finite (sometimes holonomic) sequences is [KP11]. The authors of [Ken+21] gave the following result that connects the decidability of POSITIVITY to that MINIMALITY for second-order P-finite sequences.

Theorem 12 ([Ken+21]). *Given a second-order P-finite recurrence relation of the form (4). We have the following:*

1. *There is an interval $P \subseteq \mathbb{R}$ such that all the terms of any non-trivial solution $\langle u_n \rangle_n$ are non-negative if and only if $u_0/u_{-1} \in P$.*
2. *We can decide POSITIVITY except when u_0/u_{-1} coincides with an endpoint of P .*

Further, for the class of second-order P-finite sequences, the problem of deciding POSITIVITY Turing-reduces to that of MINIMALITY (i.e., a procedure to decide MINIMALITY implies the existence of a procedure to decide POSITIVITY).

3.1 Introducing polynomial continued fractions

Example 13. The following continued fraction expansions are well-known,

$$\begin{aligned} \phi &= 1 + \mathbf{K}_{n=1}^{\infty} \frac{1}{1}, \\ \frac{4}{\pi} &= 1 + \mathbf{K}_{n=1}^{\infty} \frac{(2n-1)^2}{2}, \\ \zeta(3) &= \frac{6}{5 + \mathbf{K}_{n=1}^{\infty} \frac{-n^6}{34n^3 + 51n^2 + 27n + 5}}. \end{aligned}$$

Motivated by the continued fractions of the type featured in [Example 13](#), Bowman and MC Laughlin [\[BM02\]](#) coined the term *polynomial continued fractions* for the class of continued fractions whose partial quotients are (ultimately) polynomials.

PCF EQUALITY PROBLEM: Given a convergent polynomial continued fraction $\mathbf{K}_{n=1}^{\infty}(a_n/b_n)$, determine whether its limit f is equal to a particular algebraic number.

Corollary 14 ([\[Ken+21\]](#)). PCF EQUALITY and MINIMALITY for second-order P -finite sequences are irreducible.

Proof. Suppose that $\langle u_n \rangle_n$ satisfies (4). Define $\langle v_n \rangle_n$ by $v_n := u_n \prod_{j=0}^n p_3(j)$. Then $\langle v_n \rangle_n$ satisfies the recurrence relation

$$v_n = p_2(n)v_{n-1} + p_1(n)p_3(n-1)v_{n-2}. \quad (5)$$

Moreover, $\langle u_n \rangle_n$ is a minimal solution of (4) if and only if $\langle v_n \rangle_n$ is a minimal solution of (3.1). Consider the polynomial continued fraction $\mathbf{K}(a_n/b_n)$ associated with (3.1) has partial quotients $a_n := p_1(n)p_3(n-1)$ and $b_n := p_2(n)$. By [Theorem 11](#), $\langle v_n \rangle_n$ is a minimal solution of if and only if $\mathbf{K}(a_n/b_n) \rightarrow -v_0/v_{-1}$. Thus given an oracle to determine the value of a polynomial continued fraction, one can determine whether $\langle v_n \rangle$ is minimal. Since the transformation from $\langle u_n \rangle_n$ to $\langle v_n \rangle_n$ preserves minimality, we can determine whether $\langle u_n \rangle_n$ is minimal.

Conversely, given a convergent polynomial continued fraction $\mathbf{K}(a_n/b_n)$ and ξ a real algebraic number, construct a sequence $\langle v_n \rangle_{n=-1}^{\infty}$ as follows. Let $\langle v_n \rangle_n$ be a solution to the associated recurrence relation $v_n = b_nv_{n-1} + a_nv_{n-2}$ with initial values $v_{-1} = 1, v_0 = \xi$. By [Theorem 11](#), $\langle v_n \rangle_n$ is a minimal solution if and only if $\mathbf{K}_{n=1}^{\infty}(a_n/b_n) \rightarrow -v_0/v_{-1} = \xi$. Hence if you have an oracle for determining minimality, then one can test the value of a polynomial continued fraction. $\square$

We move onto convergence results for continued fractions. The following collects together results from many sources (a good reference is [\[LW92\]](#)).

Theorem 15 (Generalised Worpitzky). Let κ_n be a rational function in $\mathbb{Q}(n)$ and consider the continued fraction $\mathbf{K}(\kappa_n/1)$. If $\kappa_n < 0$ for all sufficiently large $n \in \mathbb{N}$, then $\mathbf{K}(\kappa_n/1)$ converges to a value in $\hat{\mathbb{R}}$ if and only if, either

1. the limit $\lim_{n \rightarrow \infty} \kappa_n$ exists and is strictly greater than $-1/4$, or
 2. $\lim_{n \rightarrow \infty} \kappa_n = -1/4$ and moreover, $\kappa_n \geq -1/4 - 1/(4n)^2 - 1/(4n \log n)^2$ for all sufficiently large n .
- Since we assume that κ_n is a rational function, the ‘eventually’ statement is effective in our setting.

Remark 16. The coefficient $-1/16$ of the $1/n^2$ term is best possible in the sense that if

$$\kappa_n = \begin{cases} -1/4 - \varepsilon_1/n^2 + O(1/n^3) & \text{with } \varepsilon_1 > 1/16, \text{ or} \\ -1/4 - \varepsilon_2/n + O(1/n^2) & \text{with } \varepsilon_2 > 0, \end{cases}$$

then $\mathbf{K}(\kappa_n/1)$ diverges. In essence, the convergence properties of $\mathbf{K}(\kappa_n/1)$ are connected to the convergence properties of $\mathbf{K}(\kappa)$ (where $\kappa_n \rightarrow \kappa$ as $n \rightarrow \infty$). For constant partial coefficients, the convergence properties are related to the associated linear fractional transformations $s(z) = \kappa/(1+z)$. The continued fractions $\mathbf{K}(\kappa_n/1)$ are termed *limit hyperbolic*, *limit parabolic*, and *limit elliptic* type and are so named for the limiting linear fractional transformation.

The second convergence condition says that a limit parabolic type continued fraction converges only if the speed of convergence of $\liminf_{n \rightarrow \infty} \kappa_n$ to $-1/4$ is sufficiently fast.

Remark 17. Given a solution $\langle u_n \rangle_n$ to (2), we can apply a transformation $\langle u_n \rangle_n \mapsto \langle w_n \rangle_n$ such that $\langle w_n \rangle_n$ satisfies the recurrence relation

$$w_n = w_{n-1} + \kappa_n w_{n-2} \quad \text{with } \kappa_n := \frac{a_n}{b_n b_{n-1}} \text{ where } a_n := \frac{p_1(n)}{p_3(n)} \text{ and } b_n := \frac{p_2(n)}{p_3(n)}. \quad (6)$$

Under the assumption that $\kappa_n < 0$, it is straightforward to show that this transformation preserves both positivity and minimality of sequences; that is to say, $\langle u_n \rangle_n$ is a minimal solution of (2) (resp. positive solution) if and only if $\langle w_n \rangle_n$ is a minimal solution of (6) (resp. positive solution).

The condition $\kappa_n < 0$ is no great imposition given our interest in POSITIVITY for second-order P-finite sequences. Indeed, POSITIVITY is inherently related to the ultimate signs of the polynomial coefficients in the recurrence relation (2) and, as seen in [Ken+21], the positivity instances where $\kappa_n > 0$ for all sufficiently large n are easy to decide. Hereafter we assume that $\kappa_n < 0$ for all sufficiently large n . To reduce cognitive burden on the reader, we typically reduce ‘sufficiently large’ or ‘eventually’ statements to the consider of problem instances where, say, $\kappa_n < 0$ holds for all n .

Taken together with the Śleszyński–Pringsheim Theorem (cf. [Ken+21; LW92]), the Generalised Worpitzky Theorem (Theorem 15) leads us to the following properties of $\mathbf{K}(\kappa_n/1)$ with $\kappa_n < 0$ for all sufficiently large n .

1. The sequence of approximants $\langle A_n/B_n \rangle_n$ of the continued fraction $\mathbf{K}(\kappa_n/1)$ strictly decreases and converges to a finite value in $\mathbb{R}$.
2. For each $n \in \mathbb{N}$, $B_n > 0$.

Under our current assumptions, we can see that determining the positivity of sequence $\langle w_n \rangle_n$ is equivalent to determining the positivity of sequence $\langle u_n \rangle_n$. For $\langle w_n \rangle_n$ to be positive, then necessarily $w_{-1}, w_0 \geq 0$. Further $w_n \geq 0$ for each $n \in \mathbb{N}$. Using the basis property of the canonical numerator and denominator,

$$w_n = w_{-1}A_n + w_0B_n \geq 0 \text{ if and only if } \frac{A_n}{B_n} \geq -\frac{w_0}{w_{-1}}.$$

Then the Śleszyński–Pringsheim Theorem says that we have the bound

$$-\frac{w_0}{w_{-1}} \leq \mathbf{K}_{n=1}^{\infty} \left(\frac{\kappa_n}{1} \right).$$

This bound, clearly phrased in terms of the sequence $\langle w_n \rangle_n$, is not *quite enough* to establish the interval result stated in Theorem 12 (stated in terms of $\langle u_n \rangle_n$), but we have covered the major steps.

We have one statement in Theorem 12 left to comment on. The work in [Ken+21] establishes a semi-algorithm that takes as inputs solution sequences $\langle u_n \rangle_n$ of (2). The semi-algorithm terminates for sequences that are dominant and correctly outputs ‘input is a positive sequence,’ or ‘input is not a positive sequence’ as appropriate. The semi-algorithm is non-terminating when given a minimal solution as an input.

Part of the underlying analysis of the above-mentioned semi-algorithm is an analysis of *tail sequences*, which are sequences of the form $\langle w_n/w_{n-1} \rangle_n$ and satisfy a non-linear first order recurrence relation

$$-\frac{w_n}{w_{n-1}} = -1 + \kappa_n \cdot \left(-\frac{w_{n-1}}{w_{n-2}} \right)^{-1}.$$

The cases where the recurrence relations relate to limit hyperbolic and limit parabolic type continued fractions are discussed in [Ken+21].

3.2 A decidability result for second-order degree-1 P-finite sequences

One of the major obstacles to a decision procedure in [Ken+21] are algorithms for performing equality tests between two real numbers.

Definition 18 ([KZ01]). A *period* is a complex number whose real and imaginary parts are values of absolutely convergent integrals of algebraic functions with algebraic coefficients over domains in $\mathbb{R}^k$ given by polynomial inequalities with algebraic coefficients

$$\int_D g(x_1, \dots, x_k) dx_1 \cdots dx_k.$$

Examples of periods include the algebraic numbers and $\pi = \int_0^\infty \frac{2}{x^2+1} dx$.

Conjecture 19 ([KZ01]). *It is decidable whether two periods are equal.*

We term the class of P-finite sequences that obey second-order recurrence relations whose polynomial coefficients are all linear, i.e., relations of the form

$$(\alpha_1 n + \alpha_0)u_n = (\beta_1 n + \beta_0)u_{n-1} + (\gamma_1 n + \gamma_0)u_{n-2} \quad (7)$$

the *second-order degree-one P-finite sequences*. Subject to further restrictions, we have the following decidability result.

Theorem 20 ([Ken+]). *POSITIVITY and MINIMALITY are decidable for the class of second-order degree-one P-finite sequences with two distinct characteristic roots (i.e., the polynomial $p(x) = \alpha_1 x^2 - \beta_1 x - \gamma_1$ associated with the recurrence relation (7) has two distinct rational roots).*

Under these restrictive conditions, the equality checks performed as part of the decision procedure make use of the following recent results:

1. An algorithm, due to Fischler and Rivoal [FR22], for determining all polynomial relations of a given family of E-functions evaluated at a common algebraic point.
2. An algorithm, due to Sertöz, Ouaknine, and Worrell [SOW25], for determining all linear relations of 1-periods.

Recall that E-functions generalise the exponential function and were introduced by Siegel, whereas 1-periods are integrals of algebraic functions of a single variable.

4 Summary

Let us summarise the content of the mini-course and suggest a few questions going forward.

Lecture 1: We discussed how common decision problems in Computer Science for simple linear loops can be posed in terms of decision problems for C-finite sequences. For C-finite sequences POSITIVITY is decidable at low-orders, but is an open decision problem for higher-order sequences.

Lecture 2: We abandoned the C-finite setting and instead focused on second-order recurrence sequences. In particular, we introduced the notions of minimal and dominant solutions to second-order recurrence relations. Through Pincherle's theorem (Theorem 11), we saw a connection to the convergence of continued fractions.

Lecture 3: We discussed POSITIVITY, MINIMALITY, and PCF EQUALITY tests for second-order P-finite sequences and the related class of polynomial continued fractions. We demonstrated recent progress on these problems in the field.

We pose the following questions.

Question 1. Are there classes of convergent continued fractions whose limits are sufficiently 'nice' so that we can circumvent the hard problem of testing equalities in the general setting? [Aside: the work of [Ken+] approaches this type of restriction from the *recurrence sequence* side, rather than the *continued fraction* side.]

Question 2. Polynomial continued fractions have 'well-behaved' partial quotients (the $\langle a_n \rangle_n$ and $\langle b_n \rangle_n$ defining $K(a_n/b_n)$). Can we extend or generalise the methods in [Ken+21; Ken+] to other classes of 'well-behaved' continued fractions.

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